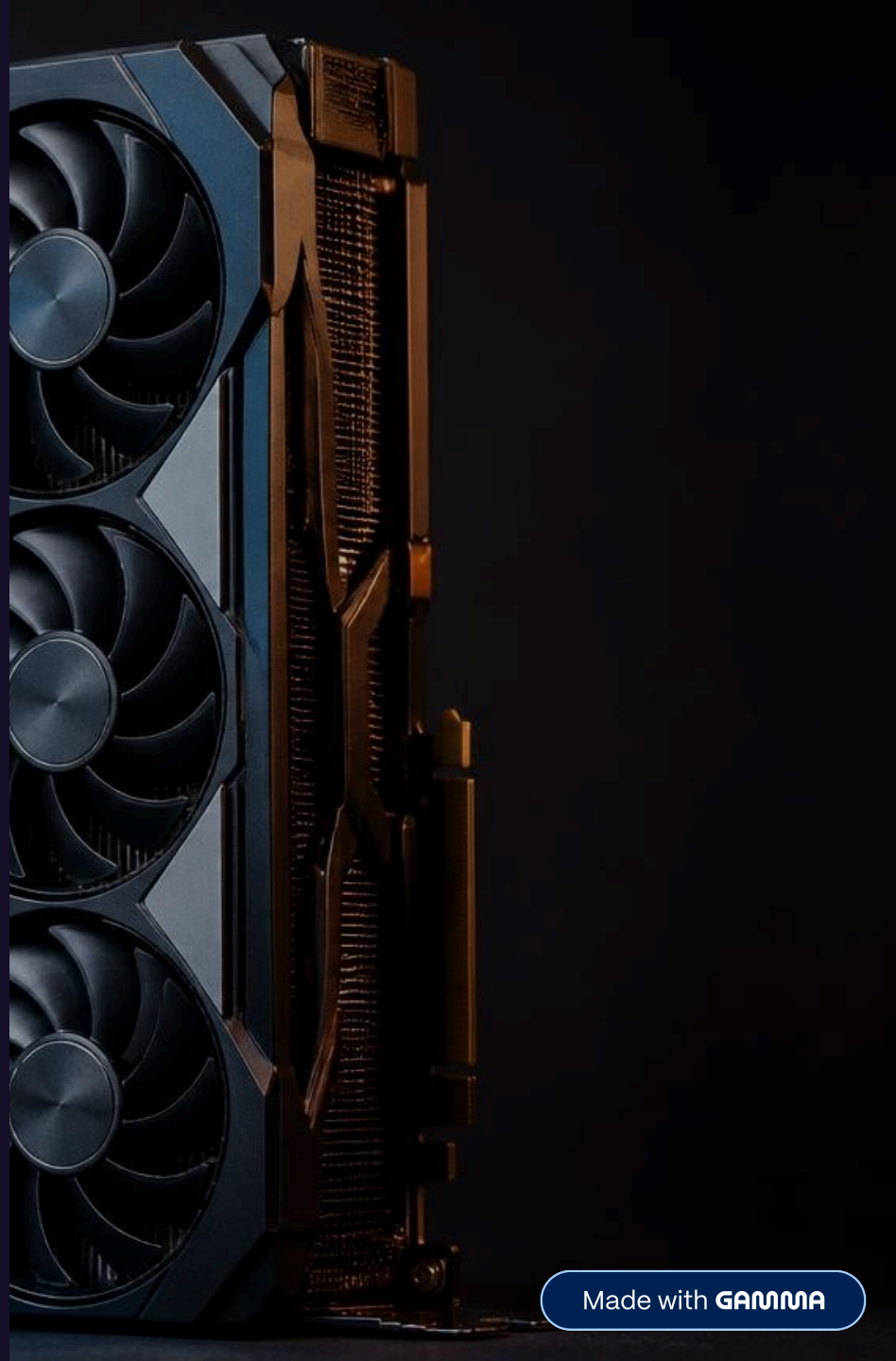
HARDWARE ASSIGNMENT

# GPU Cooling: Air vs Liquid

**How to identify the GPU, understand its cooling system, and choose a practical setup for IPL streaming and gaming.**



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START HERE

# Step 1: Identify the GPU

Before comparing cooling, record exactly what hardware is installed.

01

## Record the brand

Write the manufacturer, such as NVIDIA, AMD, or Intel.

BRAND

**NVIDIA**

02

## Record the model

Capture the full model number, not only the brand.

MODEL

**GeForce RTX 4060**

03

## Mark the type

Is it integrated or dedicated?

TYPE

**Dedicated (own VRAM)**

TWO QUICK CHECKS

Task Manager → Performance → GPU
dxdiag → Display tab

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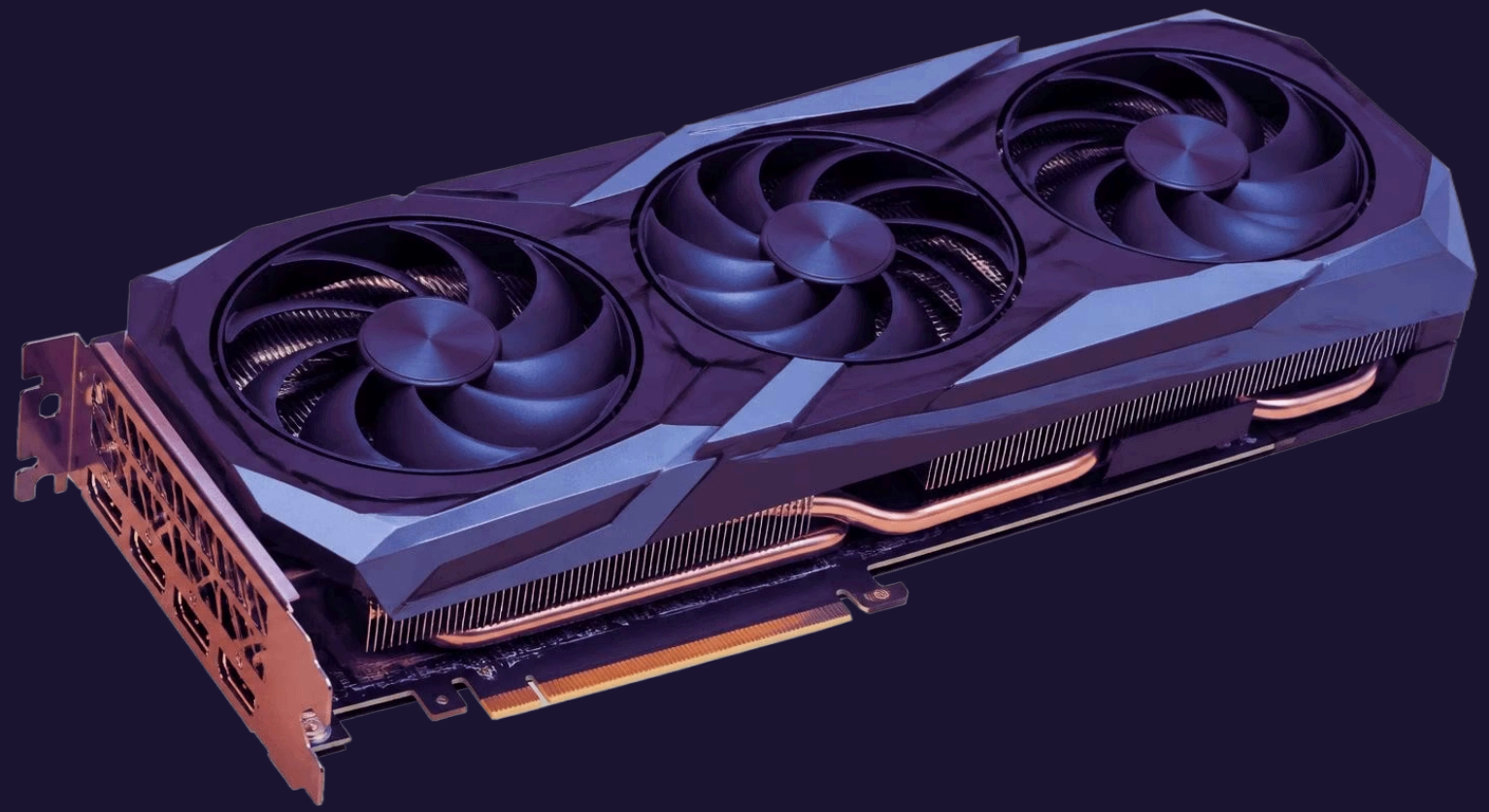
STEP 2 · AIR COOLING

# Inside the cooling system

A GPU cooling system moves heat away from the chip, spreads it through metal, and sends it into the surrounding air.

STAND-IN IMAGE

Replace the image with your own GPU photo for the final assignment.



### FAN

Pushes air across the fins.

### HEATSINK

Absorbs and spreads heat.

### HEAT PIPES

Carry heat to the fins.

### THERMAL PASTE

Fills the gap above the chip.

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LIQUID COOLING

# Liquid cooling parts

An AIO cooler moves heat through coolant before releasing it through a radiator.

## Pump block

Moves coolant across the hot chip.

## Radiator

Releases heat to the air.

## Tubing

Carries coolant between the parts.

## Radiator fans

Push air through the radiator fins.

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THE DECISION

# Air vs liquid cooling compared

| Factor | Air cooling | Liquid cooling |
| --- | --- | --- |
| **Cost** | Cheaper and simpler. | Roughly ₹8,000–20,000+ extra. |
| **Efficiency** | Works well for most gaming builds. | Removes more heat and runs quieter at full load. |
| **Maintenance** | Needs dust cleaning. | Has leak and pump risk. |
| **Installation** | Usually straightforward. | Needs radiator space and careful routing. |
| **Reliability** | Fewer failure points. | Depends on pump and sealed loop health. |

Liquid cooling can offer more thermal headroom, but the extra parts and risk need to earn their place.

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MY PICK

# Air cooling for IPL streaming and gaming

- Handles long gaming and streaming sessions.
- Costs less; savings can fund a stronger GPU.
- Low maintenance suits India’s dusty, warm conditions.

CHOOSE LIQUID ONLY FOR

Overclocking a high-end GPU



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